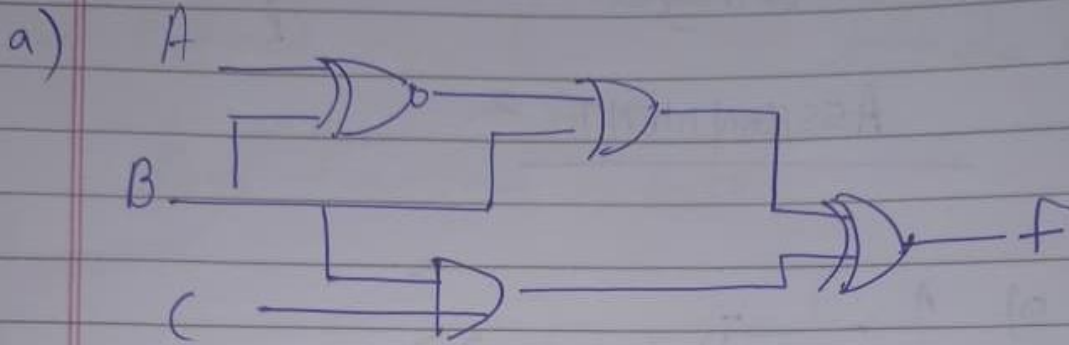


ASSIGNMENT 2



$$f = (\overline{A \oplus B}) + B \oplus BC$$

$$f = AB + \overline{A}\overline{B} + B \oplus BC$$

$$= B(A+1) + \overline{A}\overline{B} \oplus BC$$

$$= B + \overline{A}\overline{B} \oplus BC$$

$$= (B + \overline{A})(B + \overline{B}) \oplus BC$$

$$= (B + \overline{A})(1) \oplus BC$$

$$= B + \overline{A} \oplus BC$$

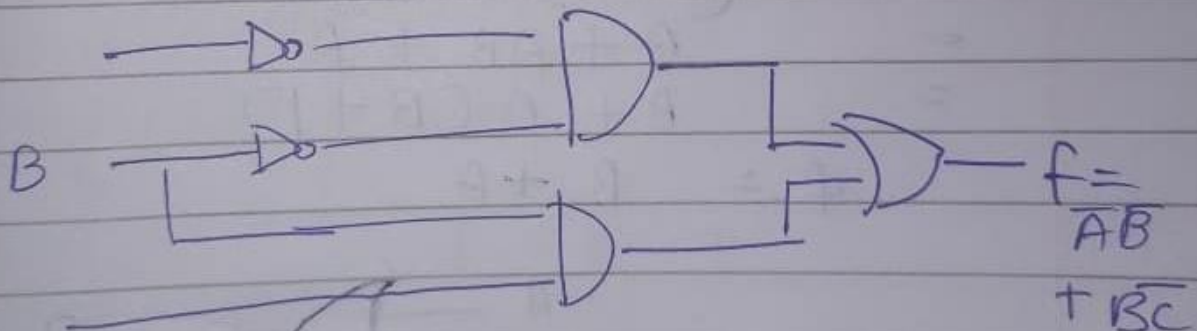
$$= (B + \overline{A})BC + (B + \overline{A})\overline{B}\overline{C}$$

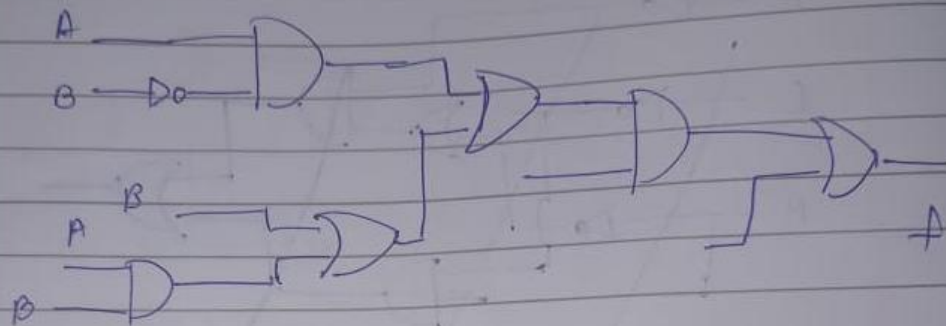
$$= (\overline{B}A)BC + (B + \overline{A})(\overline{B} + \overline{C})$$

$$= 0 + \cancel{B\overline{B}} + B\overline{C} + \overline{A}\overline{B} + \overline{A}\overline{C}$$

$\overline{A}\overline{C}$ is absorbed $\therefore$

$$f = \overline{A}\overline{B} + B\overline{C}$$





$$F = ((A\bar{B}) + (B + AB)) B + A$$

$$F = ((A\bar{B}) + B(\cancel{1+A})) B + A$$

$$= (A\bar{B} + B) B + A$$

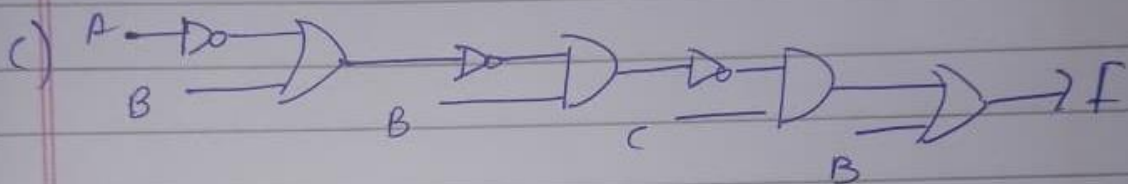
$$= [(B+A)(B+\bar{B})] B + A$$

$$= (B+A) B + A$$

$$= B + AB + A$$

$$= B + A(B+1)$$

$$F = B + A$$



$$F = \overline{\overline{(\overline{A+B})B}} C + B$$

$$F = \overline{\overline{\overline{A} \overline{B} B}} C + B$$

$$F = \underline{\underline{B}}$$

$$2a) \quad \overline{A}C\overline{D}E + \overline{A}B\overline{D} + ABCE + ABD \\ = \overline{A}\overline{D}(CE+B) + AB(CE+D)$$

$$b) \quad \overline{A}\overline{B}\overline{C} + ABD + \overline{A}C + \overline{A}C\overline{D} + A\overline{C}D + \overline{A}B\overline{C} \\ = \overline{B}\overline{C}(\overline{A} + A) + \overline{A}(C + C\overline{D}) + AD(B + \overline{C}) \\ = \overline{B}\overline{C} + \overline{A}(C(1 + \overline{D})) + AD(B + \overline{C}) \\ = \overline{B}\overline{C} + \overline{A}C + AD(B + \overline{C})$$

$$2c) \quad \overline{A}\overline{C} + \overline{B}C + \overline{A}B + \overline{A}BC + A\overline{C}\overline{D} + \overline{B}C\overline{D} \\ = \overline{B}C(1 + \overline{A}) + \overline{C}(\overline{A} + \overline{B}\overline{D}) + A(\overline{B} + BC) \\ = \overline{B}C + \overline{C}(\overline{A} + \overline{B}\overline{D}) + A(\overline{B} + C)$$

$$= \overline{B}C + \overline{C}(\overline{A} + \overline{B}\overline{D}) + A(\overline{B} + C) \\ = \overline{B}C + \overline{C}(\overline{A} + \overline{B}\overline{D}) + A(\overline{B} + C)$$

$$2d) \quad \overline{x}_1 \overline{x}_3 \overline{x}_4 + x_3 x_4 + \overline{x}_1 \overline{x}_2 x_4 \\ + x_1 x_2 \overline{x}_3 x_4 + x_2$$

$$\Rightarrow x_2(x_1 \overline{x}_3 x_4 + 1) +$$

$$\Rightarrow \overline{x}_1 \overline{x}_3 \overline{x}_4 + x_3 x_4 + \overline{x}_1 \overline{x}_2 x_4 \\ + x_2$$

$$\Rightarrow \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 x_4 + x_2$$

$$\Rightarrow \bar{x}_1 \bar{x}_3 + x_3 x_4 + \bar{x}_1 x_4 + x_2$$

$$\Rightarrow \bar{x}_1 \bar{x}_3 + x_3 x_4 + x_2$$

3.

$$F = \bar{A} \bar{B} C + \bar{A} B \bar{C} + A \bar{B} \bar{C} + A B C$$

$$= \underline{A \oplus B \oplus C}$$

4.

$$a) \sum_m (4, 6, 8, 10, 11, 12, 15) + \sum_d (3, 5, 7, 9)$$

$ab \backslash cd$	$\bar{c}\bar{d}$	$\bar{c}d$	$c\bar{d}$	cd
$\bar{a}\bar{b}$	0	1	X ₃	2
$\bar{a}b$	X ₄	X ₅	X ₇	6
ab	12	13	15	14
$a\bar{b}$	X ₈	9	X ₁₁	10

$$cd + \bar{a}b + a\bar{b} + \bar{c}\bar{d}b$$

$$= cd + \bar{c}\bar{d}b + a \oplus b$$

4b) $f(a, b, c, d) = \prod_{max}(0, 2, 4, 6, 7).$
 $\prod_{min}(10, 11)$

ab \ cd	cd			
	00	01	11	10
00	0	1	3	0
01	0	4	5	0
10		12	13	15
11	8	9	X	X

$f = (\bar{a} + \bar{d})$
 $(\bar{a} + b + c)$